communities, or patches, do not always have clear boundaries that make them recognizably separate, and that different species may respond to things happening at a different scale. They identified three kinds of metacommunity: markedly separate patches; short-lived but distinct patches that appear in a habitat from time to time at varying size; and permanent patches with vague or “blurred” boundaries.

Distinct patches

The most obvious markedly separate patches are islands in the ocean. These are a convenient subject to study and there is a vast literature on island biogeography, reaching back to Charles Darwin’s famous study of variations between finches in the Galápagos Islands in the Pacific Ocean. Neatly separate patches make good subjects for study, which is why they have been popular with community ecologists. But, of course, birds and many other organisms blown across by the wind or washed in by the sea ensure that even island communities are never completely isolated. This is why some metacommunity studies focus on the space between the communities even where the patches are distinct, as they are with ponds and lakes, and analyse how species move between them.

Short-lived but distinct patches may be much harder to identify, simply because of their ephemeral nature. Nonetheless, ecologists have made metacommunity studies of holes in trees that fill with water for a period of time following a storm, patches of fungal fruiting bodies that live just a few days or weeks, and even pitcher plants that, after dew or rain, provide a short-lived aquatic home for both bacteria and insects.